National University
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VIRTUAL STOCK TRADING SIMULATOR

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1. Introduction

This project is a Stock Market Assistant System designed to help users interact with stock market data using efficient data structures and modern APIs. The system includes functionalities such as user authentication, stock data fetching, stock recommendations, user data storage, and an AI-powered chatbot for financial queries.

The goal of this project is to apply concepts learned in Data Structures & Algorithms—such as Dynamic Arrays and Hash Maps—while integrating them into a practical, real-world application. The project also extends beyond traditional DSA by incorporating APIs and JSON/CSV parsing to handle live stock market information.

2. Project Features

2.1 User Authentication

- Users can sign up, log in, and securely store credentials.
- Passwords are safely hashed before storage.
- User data is stored in separate files under the users_data directory.

2.2 Stock Data Fetching

PSX (Pakistan Stock Exchange) Data

- Fetched using local JSON or CSV files.
- Parsed and stored using custom data structures.

Yahoo Finance Data

- Real-time stock price extraction.
- Used for updated stock recommendations.

2.3 Custom Data Structures Implemented

1. Dynamic Array
2. Hash Map
3. LinkedList
4. Stack
5. Priority Queue

2.4 Stock Recommendation System

- Analyzes stock price trends.
- Recommends stocks based on performance, volatility, and return metrics.
- Uses sorting for analysis.
- Operates using Dynamic Arrays + HashMap data retrieval.

2.5 AI Chatbot (Gemini API Integrated)

Capabilities:

- Answers financial and stock-related questions
- Provides natural, conversational responses
- Understands user intent without keywords or pattern matching

Implementation summary:

- User enters a query
- Query is sent to Gemini API
- API processes and returns an intelligent response
- Chatbot displays the output

2.6 User Data Storage

Each user has a dedicated file containing:

- Login credentials
- Stock viewing history
- Preferences

2.7 Main Menu System

Allows users to:

- Fetch stock data
- Analyze stocks (Buy and Sell)
- View recommendations
- Interact with the chatbot
- Check their saved history

3. Methodology

3.1 Dynamic Array

Where used:

- Storing stock datasets fetched from CSV/JSON
- Holding user history
- Maintaining lists of recommended stocks

Why used:

- O(1) access time for retrieving stock information
- Auto Resizing
- Efficient traversal when displaying stocks or recommendations

Problem Solved:
Stock datasets can vary in size, and a fixed-size array would either waste memory or cause overflow. The Dynamic Array provides flexibility and ensures efficient access while maintaining structure, making it ideal for storing stock prices and history logs.

3.2 Hash Map

Where used:
• Stock symbol (Key), price (Value)
• Storing parsed stock data

Why used:

- O(1) average lookup time
- Perfect for searching stock data by ticker symbol
- Avoids slow linear searches in large datasets

Problem Solved:
Stock search operations were slow when using only lists. The Hash Map enabled instant access to specific stocks, greatly improving performance when the user searches or when the system analyzes a stock during recommendation.

3. 3Linked List

Where Used:

- Managing sequences of user actions
- Handling stock updates or tracking recently viewed stocks
- Any feature requiring frequent insertions/deletions

Why Linked List?

- Frequent insertions or deletions occur
- The size of the structure changes continuously
- You want efficient memory usage and flexible data growth

Problem Solved:
Dynamic updates to user data (e.g., adding or removing recent actions) required a structure where elements could be added or removed without shifting large sections of memory.
The Linked List enables efficient O(1) insert/delete operations, making it ideal for maintaining sequences of changing data.

3.4 Stack

Where Used:

- Tracking user navigation history
- Providing undo-like functionality for chatbot/user operations
- Managing last-in, first-out (LIFO) interactions

Why Stack?

- A Stack is useful because:
- It naturally supports **LIFO** behavior
- Operations are very fast (push/pop = O(1))
- Ideal for keeping track of previous user steps

Problem Solved:
Users often need to go back to previous menus or actions. The Stack makes it possible to store the history of actions and quickly retrieve the most recent one.
This helps improve user experience and organizes the flow of interactions.

3.5. Priority Queue

Where Used:

- Ranking and selecting the best-performing stocks
- Sorting stocks based on metrics such as:
    1. Compound Annual Growth Rate (CGAR)
    2. Price Change

Why Priority Queue?

- A Priority Queue enables:
- Automatically ordering elements based on priority
- Efficient extraction of the highest priority (best stock)
- Faster selection compared to sorting entire datasets repeatedly

Problem Solved:
Stock recommendation requires picking the *best* stocks based on performance. Instead of sorting the stock list every time (which is O(n log n), the Priority Queue efficiently gives the top-performing stock(s) with O(log n) operations.
This ensures smooth and fast recommendation output.

RESULT:

```
============================================================
  STOCK INVESTMENT SYSTEM
============================================================
 Loaded 3 existing user(s)
 Loaded CAGR data for 10 stocks
 Portfolio initialized with 10 stocks


============================================================
MAIN MENU
============================================================
1.   Sign Up (New User)
2.   Log In (Existing User)
3.   Exit
============================================================
Enter your choice (1-3): 2

============================================================
  LOGIN TO YOUR ACCOUNT
============================================================
Username: AimanFarooqui
Password: ********

 Welcome back, AimanFarooqui!

Press Enter to continue...
```

1)Chatbot:

```
============================================================
 USER: AimanFarooqui
============================================================
1.  ChatBot (Stock Recommender)
2.  Dummy Trading (Learn Trading)
3.  Stock Prediction (5-Year Profit)
4.  Logout
============================================================
Enter your choice (1-4): 1
 Chatbot started! Type 'exit' to quit.

You: hi
 Bot: Hi! Ready to help you navigate the Pakistan Stock Market, understand its potential and risks, and explain key investing terms.

You: tell me about pakistan stock exchange
 Bot: Pakistan Stock Exchange (PSX) is the main stock exchange in Pakistan, where companies' shares are bought and sold, reflecting the country's economic health. Inv
esting in PSX offers potential for high returns but also carries risk due to market volatility and economic factors.

You: can you guide me how to start with this as i am new to stock exchange things
 Bot: Welcome to the Pakistan Stock Exchange (PSX)! Start by learning the basics of investing, opening a brokerage account, and researching companies before making an
y trades.

You: so how to start innvesting like what should we keep in mind
 Bot: Investing in the Pakistan Stock Exchange (PSX) involves opening a brokerage account, researching companies, and understanding market risks. Start with small, di
versified investments and prioritize long-term growth.

You: exit
 Goodbye!
```

2)Dummy Trading:

```
============================================================
 USER: AimanFarooqui
============================================================
1.  ChatBot (Stock Recommender)
2.  Dummy Trading (Learn Trading)
3.  Stock Prediction (5-Year Profit)
4.  Logout
============================================================
Enter your choice (1-4): 2

 Fetching latest stock prices...

 Fetching real-time data for 10 stocks...
   [1/10] OGDC...
   [2/10] HBL...
   [3/10] PSO...
   [4/10] LUCK...
   [5/10] MCB...
   [6/10] MARI...
   [7/10] TRG...
   [8/10] SYS...
   [9/10] SEARL...
   [10/10] FFC...
 Successfully fetched 10/10 stocks


============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 1
```

3)View Stocks:

```
============================================================
Choose an option (1-10): 1

================================================================================
▯  AVAILABLE STOCKS FOR TRADING
================================================================================
#    Symbol     Sector              Price (PKR)       Change Status
--------------------------------------------------------------------------------
1    OGDC       Oil & Gas               269.34     ▯ +0.73%  Available
2    HBL        Banking                 308.17     ▯ +0.02%  Available
3    PSO        Oil Marketing           456.13     ▯ +0.05%  Available
4    LUCK       Cement                  471.13     ▯  -1.17%  Available
5    MCB        Banking                 360.32     ▯  -0.45%  Available
6    MARI       Oil & Gas               707.52     ▯  -0.40%  Available
7    TRG        Technology               77.09     ▯ +6.65%  Available
8    SYS        Technology              151.99     ▯  -1.25%  Available
9    SEARL      Textiles                104.51     ▯ +0.62%  Available
10   FFC        Fertilizer              566.28     ▯  -2.38%  Available
================================================================================
 10/10 stocks available for trading
================================================================================

▯  Press Enter to continue...

------------------------------------------------------------
```

4) Buy Stock

```
Choose an option (1-10): 3

================================================================================
▯  AVAILABLE STOCKS FOR TRADING
================================================================================
#    Symbol     Sector              Price (PKR)       Change Status
--------------------------------------------------------------------------------
1    OGDC       Oil & Gas               269.34     ▯ +0.73%  Available
2    HBL        Banking                 308.17     ▯ +0.02%  Available
3    PSO        Oil Marketing           456.13     ▯ +0.05%  Available
4    LUCK       Cement                  471.13     ▯  -1.17%  Available
5    MCB        Banking                 360.32     ▯  -0.45%  Available
6    MARI       Oil & Gas               707.52     ▯  -0.40%  Available
7    TRG        Technology               77.09     ▯ +6.65%  Available
8    SYS        Technology              151.99     ▯  -1.25%  Available
9    SEARL      Textiles                104.51     ▯ +0.62%  Available
10   FFC        Fertilizer              566.28     ▯  -2.38%  Available
================================================================================
 10/10 stocks available for trading
================================================================================

▯  Enter stock symbol (e.g., OGDC): mcb

============================================================
▯  Stock Details: MCB
============================================================
Current Price: PKR 360.32
Change: -0.45%
Sector: Banking
Your Balance: PKR 10,000.00
Max Affordable: 27 shares
============================================================
Enter quantity to buy: 5

▯  Total Cost: PKR 1,801.60
Confirm purchase? (y/n): y
Bought 5 shares of MCB for PKR 1,801.60

▯  Press Enter to continue...
```

5)Sell Stock(when specific stock not purchased)

```
============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 4

============================================================
  YOUR CURRENT HOLDINGS
============================================================
  MCB: 5 shares @ PKR 360.32 = PKR 1,801.60
============================================================

  Enter stock symbol to sell: pso
 You don't own any PSO shares!
  Press Enter to continue...
```

When Stock has already been purchased

```
============================================================
Choose an option (1-10): 4

============================================================
  YOUR CURRENT HOLDINGS
============================================================
  MCB: 5 shares @ PKR 360.32 = PKR 1,801.60
============================================================

  Enter stock symbol to sell: mcb

============================================================
  Stock Details: MCB
============================================================
Shares Owned: 5
Current Price: PKR 360.32
Total Value: PKR 1,801.60
============================================================
Enter quantity to sell (max 5): 2

  You will receive: PKR 720.64
Confirm sale? (y/n): y
Sold 2 shares of MCB for PKR 720.64
```

6)Depositing Money

```
============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 5
ℹ  Enter amount to deposit: 20000
 Deposited PKR 20,000.00

ℹ  Press Enter to continue...
```

7)Withdraw Money

```
============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 6

ℹ  Current Balance: PKR 28,919.04
Enter amount to withdraw: 20000
Withdrew PKR 20,000.00

ℹ  Press Enter to continue...
```

8)Viewing my Portfolio

```
============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 9

============================================================
 AimanFarooqui's Portfolio
============================================================
 Balance: PKR 8,919.04
 Holdings:
   MCB: 3 shares
============================================================

  Portfolio Value Breakdown:
============================================================
  MCB: 3 × PKR 360.32 = PKR 1,080.96
============================================================
  Total Portfolio Value: PKR 10,000.00
============================================================

  Press Enter to continue...
```

9)Undo Last Action

```
============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 7
Undo: withdraw

  Press Enter to continue...

============================================================
```

After Reversing the last Action(UNDO):

```
============================================================
 DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 9

============================================================
 AimanFarooqui's Portfolio
============================================================
 Balance: PKR 28,919.04
 Holdings:
   MCB: 3 shares
============================================================

▫  Portfolio Value Breakdown:
============================================================
  MCB: 3 × PKR 360.32 = PKR 1,080.96
============================================================
▫  Total Portfolio Value: PKR 30,000.00
============================================================
```

10)REDO

Bought 10 mcb shares

```
4    LUCK       Cement                  471.13     ▫  -1.17% ▫ Available
5    MCB        Banking                 360.32     ▫  -0.45% ▫ Available
6    MARI       Oil & Gas               707.52     ▫  -0.40% ▫ Available
7    TRG        Technology               77.09     ▫ +6.65% ▫ Available
8    SYS        Technology              151.99     ▫  -1.25% ▫ Available
9    SEARL      Textiles                104.51     ▫ +0.62% ▫ Available
10   FFC        Fertilizer              566.28     ▫  -2.38% ▫ Available
==============================================================================
▫  10/10 stocks available for trading
==============================================================================

▫  Enter stock symbol (e.g., OGDC): mcb

============================================================
▫  Stock Details: MCB
============================================================
Current Price: PKR 360.32
Change: -0.45%
Sector: Banking
Your Balance: PKR 10,000.00
Max Affordable: 27 shares
============================================================
Enter quantity to buy: 10

▫  Total Cost: PKR 3,603.20
Confirm purchase? (y/n): y
▫ Bought 10 shares of MCB for PKR 3,603.20

▫  Press Enter to continue...
```

Undo the last action:

```
Press Enter to continue...

============================================================
DUMMY TRADING MENU
============================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
============================================================
Choose an option (1-10): 7
Undo: buy

Press Enter to continue...

============================================================
```

VIEWING PORTFOLIO (NOW NO STOCK BECAUSE OF UNDO)

```
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
==============================================
Choose an option (1-10): 9

==============================================
AimanFarooqui's Portfolio
==============================================
Balance: PKR 10,000.00
Holdings:
(No holdings)
==============================================

Portfolio Value Breakdown:
==============================================
==============================================
Total Portfolio Value: PKR 10,000.00
==============================================
```

REDO (BUYING 10 SHARES AGAIN)

```
=====================================================
  DUMMY TRADING MENU
=====================================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
=====================================================
Choose an option (1-10): 8
  Redo: buy

  Press Enter to continue...
```

PORTFOLIO

```
=============================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
=============================================
Choose an option (1-10): 9


=============================================
  AimanFarooqui's Portfolio
=============================================
  Balance: PKR 6,396.80
  Holdings:
  MCB: 10 shares
=============================================

  Portfolio Value Breakdown:
=============================================
  MCB: 10 x PKR 360.32 = PKR 3,603.20
=============================================
  Total Portfolio Value: PKR 10,000.00
```

STOCK PREDICTION

```
Successfully fetched 10/10 stocks


 Enter investment amount (PKR): 1200000

 Analyzing stocks using Priority Queue (Max Heap)...
=========================================================
OGDC: Expected profit = 7,370,450 PKR
HBL: Expected profit = 12,621,029 PKR
PSO: Expected profit = 10,815,360 PKR
LUCK: Expected profit = -978,109 PKR
MCB: Expected profit = 5,839,718 PKR
MARI: Expected profit = -1,136,093 PKR
TRG: Expected profit = -496,224 PKR
SYS: Expected profit = -1,136,010 PKR
SEARL: Expected profit = 4,996,481 PKR
FFC: Expected profit = 75,519,042 PKR
=========================================================
 Priority Queue built with 10 stocks
 Extracting top 3 recommendations...


=========================================================
TOP 3 RECOMMENDED STOCKS
=========================================================

 RANK 1: FFC
   Current Price:            566.28 PKR
   CAGR:                     129.69%
   Predicted 5Y Price:    36,205.28 PKR
   Profit per Share:      35,639.00 PKR
   Shares Affordable:         2,119
   Total Profit:       75,519,041.64 PKR
---------------------------------------------------------

 RANK 2: HBL
   Current Price:            308.17 PKR
   CAGR:                      63.04%
   Predicted 5Y Price:     3,550.15 PKR
   Profit per Share:       3,241.98 PKR
   Shares Affordable:         3,893
   Total Profit:       12,621,029.17 PKR
---------------------------------------------------------
```

```
----------------------------------------------------------------------

   RANK 3: PSO
   Current Price:            456.13 PKR
   CAGR:                      58.54%
   Predicted 5Y Price:     4,568.43 PKR
   Profit per Share:       4,112.30 PKR
   Shares Affordable:         2,630
   Total Profit:       10,815,359.55 PKR
----------------------------------------------------------------------

   Combined Profit: 98,955,430.37 PKR
   ROI: 8246.3%
======================================================================

   Disclaimer: Based on historical data.

============================================================
   USER: AimanFarooqui
============================================================
1.   ChatBot (Stock Recommender)
2.   Dummy Trading (Learn Trading)
3.   Stock Prediction (5-Year Profit)
4.   Logout
============================================================
Enter your choice (1-4): _
```

SIGNING OUT

```
   DUMMY TRADING MENU
=====================================
1.  View Available Stocks
2.  Refresh Stock Prices
3.  Buy Stock
4.  Sell Stock
5.  Deposit Money
6.  Withdraw Money
7.  Undo Last Action
8.  Redo Last Action
9.  Show My Portfolio
10. Back to Main Menu
=====================================
Choose an option (1-10): 10

 Returning to main menu...

=====================================
   USER: AimanFarooqui
=====================================
1.   ChatBot (Stock Recommender)
2.   Dummy Trading (Learn Trading)
3.   Stock Prediction (5-Year Profit
4.   Logout
=====================================
Enter your choice (1-4): 4

   Goodbye, AimanFarooqui!
```

CHALLENGES FACED:

1. API Integration Issues: Handling API errors from Gemini
2. Network delays fetching real-time stock data
3. Data Structure Implementation:
    - Creating a Dynamic Array from scratch
    - Handling collisions in Hash Map
    - Redo last action in dummy trading.
    - Ensuring efficient lookup times
4. Handling inconsistencies in stock symbols
5. Password masking
6. Debugging Across Multiple Modules:With many interconnected files, module dependency bugs appeared during early phases.

FUTURE IMPROVEMENT:

1. **Smarter Chatbot Integration**
   We can allow chatbot to:
    - Access user portfolio
    - Provide personalized insights
    - Suggest buy/sell strategies using real data
    - Add conversation memory so the chatbot can maintain context across multiple queries
2. Allow users to buy more than just 10 stocks:
   In the future, the system can support trading and managing multiple stocks at the same time instead of limiting users to only 10.
3. Use long-term historical data (10–15 years):
   The simulator can analyze a much larger time period of stock prices. This will help generate more accurate predictions and trends.
4. Include stock index data and market changes:
   We can incorporate information like market indexes (e.g., KSE-100), daily market changes, and overall market performance to give a better analysis.
5. Consider country economy and foreign affairs:
   The system can use economic indicators (inflation, GDP, interest rates) and major international events because these factors strongly affect stock prices.
6. Analyze company financial health:
   We can also add data about a company's revenue, profit, debt, and financial stability. This helps determine whether buying a stock is safe or risky.